

HiveForce Labs

THREAT ADVISORY



ATTACK REPORT

NailaoLocker Ransomware: Basic Design, Deadly Reach

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February 21, 2025

Last Update Date

July 28, 2025

Admiralty Code

A1

TA Number

TA2025054

Summary

Attack Commenced: June 2024

Campaign: Green Nailao

Malware: NailaoLocker Ransomware, Shadowpad, PlugX

Affected Platform: Microsoft Windows

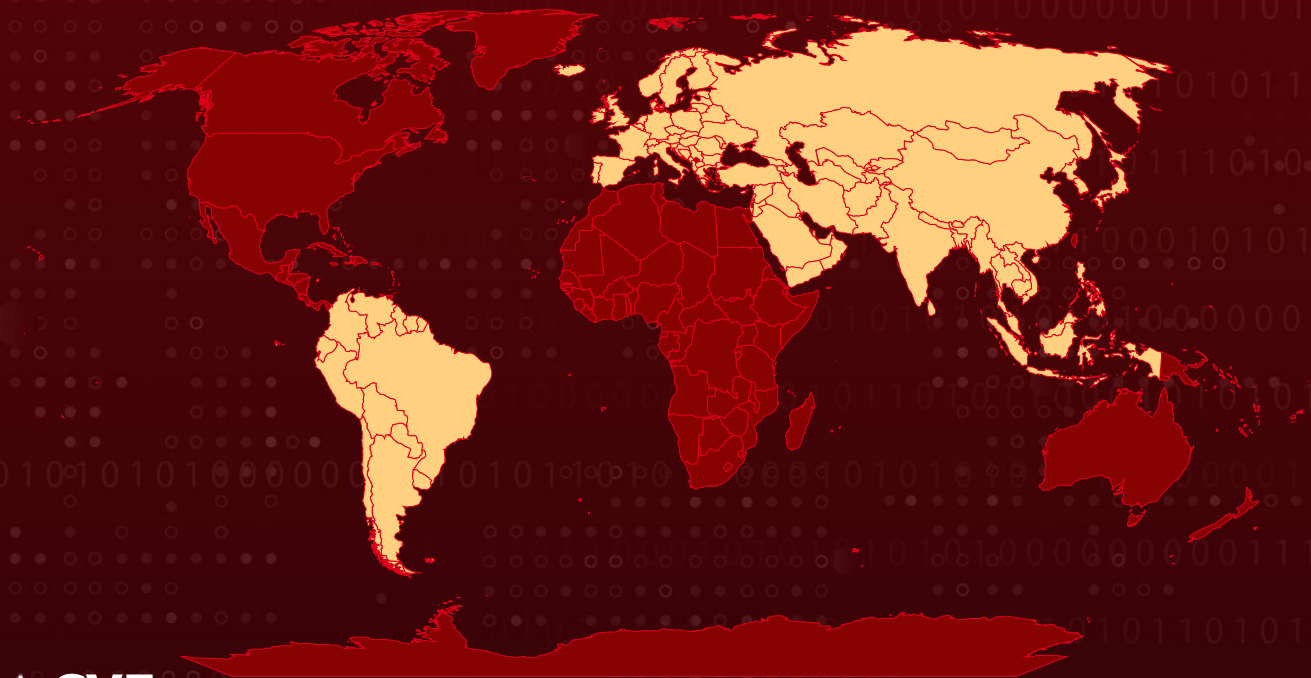
Targeted Industries: Manufacturing, Transportation, Publishing, Energy, Pharmacy, Banking, Mining, Education, Entertainment, Healthcare

Targeted Regions: Europe, the Middle East, Asia, South America

Attack: A new ransomware threat, NailaoLocker, has surfaced, making waves in the cyber world with its unexpected simplicity and dangerous associations. Exploiting a Check Point Security Gateway vulnerability, this basic yet effective malware infiltrates networks, teaming up with notorious tools like ShadowPad and PlugX, often linked to Chinese state-sponsored hackers.



Attack Regions



CVE

CVE	NAME	AFFECTED PRODUCT	ZERO-DAY	CISA KEV	PATCH
CVE-2024-24919	Check Point Security Gateway Information Disclosure Vulnerability	Check Point Security Gateway	✓	✓	✓

Powered by Bing
Tom, Zenrin



Attack Details

#1

A newly discovered ransomware strain, NailaoLocker, has emerged in recent cyberattacks, exploiting CVE-2024-24919, a vulnerability in Check Point Security Gateway to infiltrate targeted networks. Once inside, attackers deploy ShadowPad and PlugX, two notorious malware families long associated with Chinese state-sponsored threat groups.

#2

Despite its malicious intent, NailaoLocker is relatively rudimentary. Written in C++, it lacks the sophisticated features often seen in modern ransomware. It doesn't terminate security processes, evade sandbox environments, or scan for network shares. This simplistic design suggests that full encryption wasn't its primary goal.

#3

The infection process begins with DLL sideloading, where the ransomware piggybacks on a legitimate, signed executable. This executable acts as a loader dubbed NailaoLoader which first validates the environment through memory address checks. Once verified, it decrypts the main payload and injects it into memory.

#4

NailaoLocker then activates its encryption routine, using AES-256-CTR to lock files and append the ".locked" extension. After completing the encryption, it drops an HTML ransom note notably with an unusually long filename demanding payment for decryption.

#5

What makes this campaign particularly concerning is the integration of ShadowPad, a modular backdoor that has been in exclusive use by Chinese espionage actors since at least 2015. In this case, ShadowPad was used post-exploitation to bypass weak passwords and multi-factor authentication, paving the way for NailaoLocker's deployment. Green Nailao is a targeted cyber campaign observed between June and October 2024, primarily impacting European organizations, with a particular focus on the healthcare sector.

#6

The scope of the attack is significant. Threat actors have reportedly targeted 21 companies across 15 countries, spanning nine industries. This wide-reaching campaign highlights the evolving tactics of state-linked cyber groups and the growing risks posed by even unsophisticated ransomware.

Recommendations



Implement the 3-2-1 Backup Rule: Maintain three total copies of your data, with two backups stored on different devices and one backup, kept offsite or in the cloud. This ensures redundancy and protects against data loss from ransomware attacks.



Regularly Test Backup Restores: Conduct frequent tests to verify the integrity of backup data and ensure that restoration processes work as intended. This practice helps identify any issues before an actual data recovery scenario arises.



Patch Known Vulnerabilities Immediately: NailaoLocker exploits CVE-2024-24919, a known vulnerability in Check Point Security Gateway, to gain initial access. Additionally, reset passwords for both local VPN accounts and the Check Point LDAP user integrated with the gateway to mitigate potential credential compromise.



Network Segmentation & Zero Trust Implementation: Segment critical infrastructure to isolate sensitive data and limit lateral movement. Implement Zero Trust Network Access (ZTNA) by enforcing identity-based policies rather than traditional perimeter security.



Conduct Ransomware Simulation Drills: Test the organization's resilience against ransomware attacks by conducting simulated scenarios to identify gaps in preparedness.



Potential MITRE ATT&CK TTPs

<u>TA0001</u> Initial Access	<u>TA0002</u> Execution	<u>TA0003</u> Persistence	<u>TA0005</u> Defense Evasion
<u>TA0007</u> Discovery	<u>TA0040</u> Impact	<u>TA0006</u> Credential Access	<u>TA0009</u> Collection
<u>TA0010</u> Exfiltration	<u>TA0011</u> Command and Control	<u>T1190</u> Exploit Public-Facing Application	<u>T1574.002</u> DLL Side-Loading

<u>T1574</u> Hijack Execution Flow	<u>T1204</u> User Execution	<u>T1204.002</u> Malicious File	<u>T1055.012</u> Process Hollowing
<u>T1140</u> Deobfuscate/Decode Files or Information	<u>T1083</u> File and Directory Discovery	<u>T1486</u> Data Encrypted for Impact	<u>T1555</u> Credentials from Password Stores
<u>T1560</u> Archive Collected Data	<u>T1560.001</u> Archive via Utility	<u>T1041</u> Exfiltration Over C2 Channel	<u>T1055</u> Process Injection
<u>T1059</u> Command and Scripting Interpreter	<u>T1027</u> Obfuscated Files or Information	<u>T1212</u> Exploitation for Credential Access	<u>T1105</u> Ingress Tool Transfer

✂ Indicators of Compromise (IOCs)

TYPE	VALUE
Domain	updata[.]dsquarey[.]com, time[.]dsquarey[.]com, dscry[.]chtq[.]net, system[.]chtq[.]net, updata[.]chtq[.]net, network[.]oossafe[.]com, notes[.]oossafe[.]com, caba[.]superdasqe[.]me, ccs[.]superdasqe[.]me, czs[.]superdasqe[.]me, kzb[.]superdasqe[.]me, api[.]emazemedia[.]com, home[.]boopainc[.]com
Filename	logexts.dll, sensapi.dll, nView64.dll, CmLOC.dll, aaaaLOC.dll, SentinelAgentCore.dll
File Paths	C:\PerfLogs\ C:\ProgramData\Microsoft\Windows\Caches\ C:\Program Files\Microsoft\Crypto\DSS\ C:\ProgramData\Microsoft\Network\Connections\
IPv4	192[.]142[.]18[.]42,



TYPE	VALUE
IPv4	91[.]149[.]241[.]103, 208[.]85[.]16[.]252, 37[.]120[.]239[.]33, 193[.]56[.]255[.]214, 139[.]84[.]137[.]63, 158[.]247[.]199[.]185, 23[.]227[.]203[.]181, 104[.]238[.]135[.]232, 141[.]164[.]35[.]65, 176[.]222[.]55[.]131, 45[.]76[.]209[.]205, 45[.]77[.]153[.]108, 45[.]77[.]170[.]188, 47[.]242[.]0[.]122, 52[.]194[.]253[.]134, 64[.]176[.]226[.]182, 64[.]176[.]59[.]232, 64[.]176[.]65[.]49, 8[.]210[.]30[.]189, 8[.]218[.]244[.]117
MD5	a51f6c0144f4f66c40d9d18d98922ffb, 4b53171013e7b3a3e50b3ee6cc9f8ce7, 7d74814738bdc045dc7ee03669b5e949, bc267a3b45d83a0feb4d7162e6ff8113, daf90e0bc827a0d8da088c1d4c294710, 0023d66a63eb0e716d0c2dafd5f17f5f, ebf4fa09a1805297c6e272da3acf91d1, 467fedf39009b426f4abf4470a014c15, 41085ff38d803109b4da6ce46bb1c3a1, 9037a12be2e178193eef56370b63a270, c3b565e1207c7225607e20a871aad55f, 4cada74f6530e3c21374f25a3617ad8f, 027c8a7c9d892efe2301444685352b9d, e6d269115922e627309e6a06750fd518, 41d6954c06789182caf3f439977300a4, 2ae8b0f939460eacc93cecb11484da3d, 6bb2e0e349d477141dc382b68b64e351, d715581eb387e336f35c407cf6304599, 531159190bf2094995fb6616df5e68cc, 347b80290da3d6bc496b36f7c9a610ef
SHA256	8d44f2f442ca8f2fbbf75086a6f8d518c300ca93fe9957a9716076919 b475865, 83c1a668ab06f55e6879593ca24eed9f78832be97ac90bb74ef5828 067f2d900,

TYPE	VALUE
SHA256	c19be7a006bd2ba8deb56dcc6127a76f9624c6f1392a1794870dbed 6f1a81bd5, c4db25ab55af2e943a297a5ecf7a62acc3ad8897ec8ba4ab3226a13 8da237b82, 28e6362ecf033b2a26c7457dcbd7ad2ab34e253fb08666d39073391 a1254ea41, 7416f6b69b34b3a36a86e50808e1dc47f4dc665bfd6f394cef65e0ba 5eaf961b, bc490047fe6e0b0000c6cd147d3cf483105c92cf00450bfe35ac70f27 6a9e5c8, c5f8a256d0969e253633160b9728b6c2bc044f536e92af178a05a59 8aaa09c1f, a2bb321d41b2300e80f9400950fa2125470d5b3927933ab4d6397f0 cbf81532a, d74b6b2129936377aacc619bcfd4df4ffbe2f35f960a4b043b23ae78 a31ec35, 366ea3377eae28b655b530710c03fb2ace67bb531b1820e916cb0 2023892ba, f8915c5be0649642dac22572355f1462972f5087471f66f6a243f237 4b208eb8, b38dab1ee402f731313d697d5d79372ae97fcab5704077771b5b82 e705e0cd6d, 625ed0e0ad7d3fbf2738349c767a7990c9f0d388de66104e11df3e0 c4632033c, 431a630983cd327fc70ea49b3a5497a179dbde19d8f13d2cfceef4e4 7613024b, e1d72b0cfc3342b8a6436e3047c3cc54246c346ac179e459d07620d 192ba6e01, fa7f2ddf91980d639a87465bd2a38eaa44d6079b11ace3b2b3dff03c aed66de5, b28bc39e569aa0cfe984c341830cb037c5305877ba22a940c3bdaeb 43ca87878, 571607c7f55c3616e4c58db15e3d55317da10294dbc10e0cd1ed24 879b8fc051, bc5b2ef81593095696433877cccb0ab75ef942258ef4795de5538df8 42d952f4, fa3a3351cd55089d40a7311e4bfaf15e4247416f78383d94ad58809 467429b3e, 2df4c7bfa608ca88d9d659358894226910850ac0d7e566c6c10ec27 27361d47b, b6660dfe1ce69f706aaa412fcd3ff18554d604df59c09adc2a811741 7967ce9, 7b8ea6b1e2a29190cb28fc98ef837bf4a7a0b71b84177ce9395a511 3a843c4d3, de4bb30e400f081601d4091206ba6c04ac502f50e0dbac879db8c02 02bff8108,

TYPE	VALUE
SHA256	5dc36e687a7fa3cfbf845e8a53173f37ac38559b6b87f9dcf609a72b3f284035, 37039a761114251f4556e4fe41c3ec01b7206a483c4698ffe5a0f1617a8bc26b, fcb8bf42d852526214578ab4b477b29f2412a7a931c6353db4fa6c221661edf4, ceac8b67f19d596b2c2f34d682f88c717d11dd4c1144e2e7439b6bb78adb1736, 9df4624f815d9b04d31d9b156f7debfd450718336eb0b75100d02cb45d47bd9a, 28d78e52420906794e4059a603fa9f22d5d6e4479d91e9046a97318c83998679, bdf019bc6cfb239f0beae4275246216cd8ae8116695657a324497ec96e538aac, 41128b82fa12379034b3c42bdecf8e3b435089f19a5d57726a2a784c25e9d91f, c8268641aecad7bd32d20432da49bb8bfc9fe7391b92b5b06352e7f4c93bc19e, e06710652fa3c8b45fd0fece3b59e7614ad59a9bc0c570f4721aee3293ecd2d1, f4e8841a14aa38352692340729c3ed6909d7521dd777518f12b8bd2d15ea00c5, aa1233393dded792b74e334c50849c477c4b86838b32ef45d6ab0dc36b4511e3, 1248c4b352b9b1325ef97435bd38b2f02d21e2c6d494a2218ee363d9874b7607, 46f3029fcc7e2a12253c0cc65e5c58b5f1296df1e364878b178027ab26562d68, 60133376a7c8e051da787187761e596ce9b3d0cfcea21ed8f434992aa7cb8605, 0a749474b5f4a8537e50ea5b60d8c94f5c688fe414cd400c3397adca4315a509, 697e6454d9be19f0bd60aeffa0238498a91d1ea5a23112f7c8f981afd2fedb23, 7a0503da293da51a95aab0b1aa0970c8f82f04cb5149abe98fef934ba991064e, 2b069dcde43b874441f66d8888dcf6c24b451d648c8c265dff81c7dffa9d667, 27b313243daf145c9105f5372e01f1cea74c62697195c1a21c660be5f7ee788c, a2e937d0b9d5afa5b638cd511807e0fcb44ec81b354e2cf0c406f19e5564e54e, e3a669cbc8ed0cb2cf7b36d51fb747a4a5513806a3e7d0f2b866dbc07d78a690, 89ce184854e230c4cdc49d4ddd79ca688fa515029f63bc90039c983610ffec44

Patch Link

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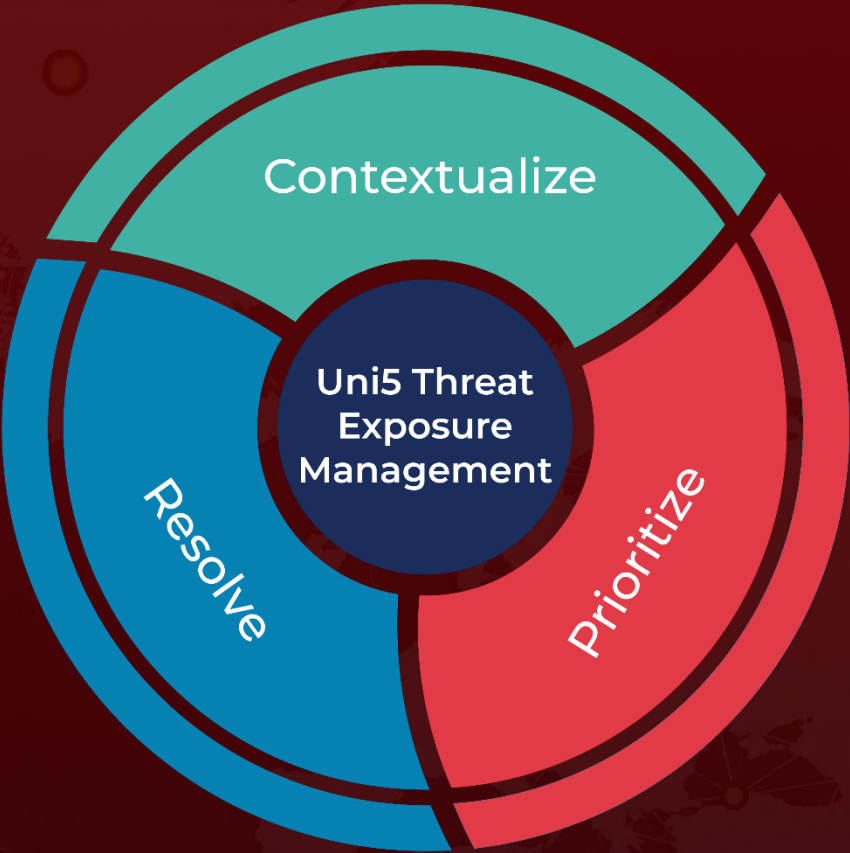
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